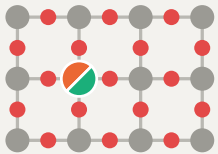
# Pr and Rb lower the O-vacancy cost; only Rb makes it spontaneous

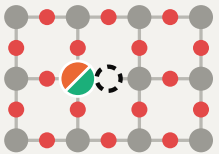
Anatase $TiO_2$, PBE+U, neutral $V_O$, O-rich limit $\mu_O = E(O_2)/2$

## ❶ Dope



Ti → $Pr^{3+}$ or $Rb^+$;
1 per cell,
48 atoms

## ❷ Remove O



Neutral $V_O$ releases ●● 2 $e^-$

## ❸ Protocol

U(Ti-3d) 3.5 eV
Γ-only, 40 Ry

✓ k-mesh, cut-off, D3: Pr–Rb Δ shifts ≤ 0.04 eV

! smearing: Pr cells polarise, Pr +0.99 eV

## ❹ Bookkeeping

○ $h^+$ ● $e^-$ (formal)

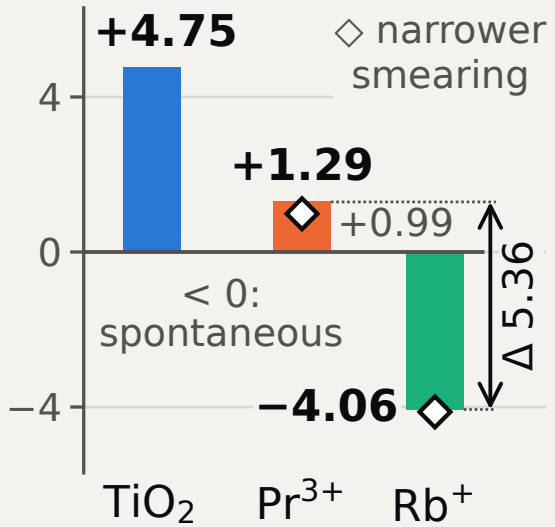
● $Pr^{3+}$

○ + ●● → ●

1 $e^-$ left: on Ti-3d

● $Rb^+$

○○○ + ●● → ○

1 $h^+$ left: on O-2p

## ❺ Result: $E_f$ (eV)